User Referral Program

Cleaned-up solution and recommendations

Executive Summary

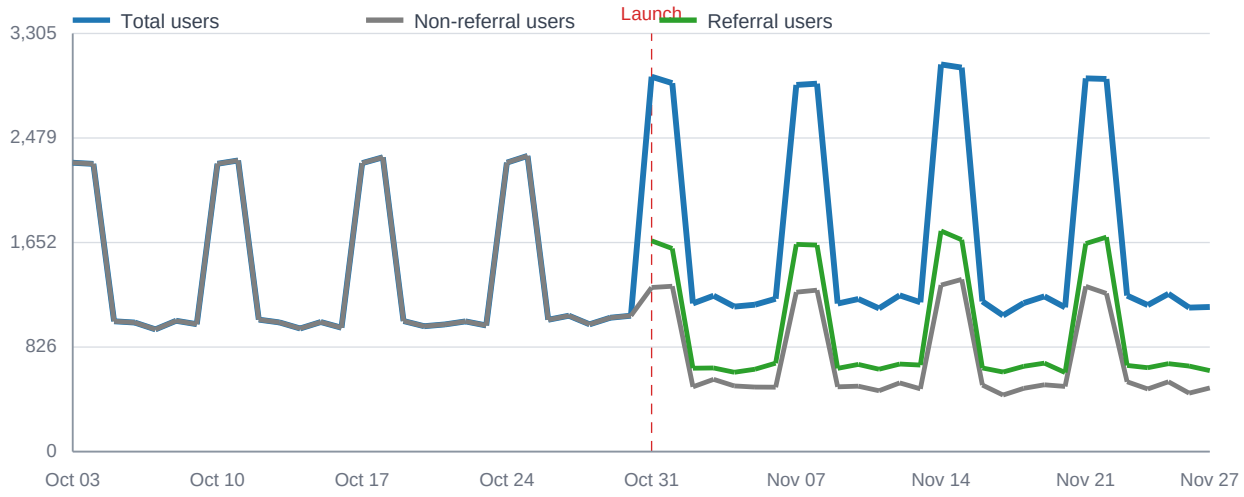
Company XYZ launched a referral program on October 31, offering a $10 credit when a referred user makes a purchase. The data covers 97,341 purchase transactions across a balanced 28-day pre-launch and 28-day post-launch window.

After launch, average daily revenue increased by 16.8% and average daily purchasing users increased by 21.8%. Total revenue rose by $337,607. The launch therefore looks directionally positive, but the data also shows signs of cannibalization and possible referral abuse.

Metric	Before launch	After launch	Change
Transactions	47,341	50,000	+2,659
Unique purchasing users	5,000	18,397	+13,397
Revenue	$2,006,396	$2,344,003	$337,607
Average order value	$42.38	$46.88	$4.50

What Changed After Launch

Daily Purchasing Users

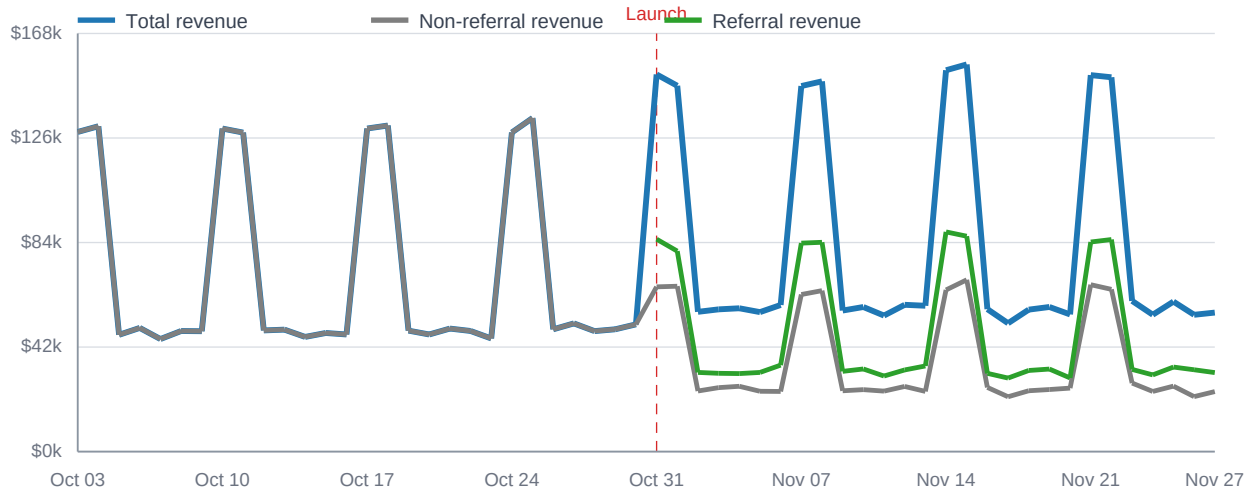


The user count rises after launch, but the mix changes sharply. Once the inconsistent referral labels are corrected at the user level, daily non-referral users fall while referred users appear immediately.

Daily average	Before launch	After launch
Total purchasing users	1,384	1,687
Non-referral users	1,384	737
Referral users	0	950

Revenue Impact

Daily Revenue



Average daily revenue increased from $71,657 before launch to $83,714 after launch. The lift is real in the observed data, but it should not be interpreted as causal proof because there was no holdout group.

Daily average revenue	Before launch	After launch
Total revenue	$71,657	$83,714
Non-referral revenue	$71,657	$36,521
Referral revenue	$0	$47,193

Referral Contribution

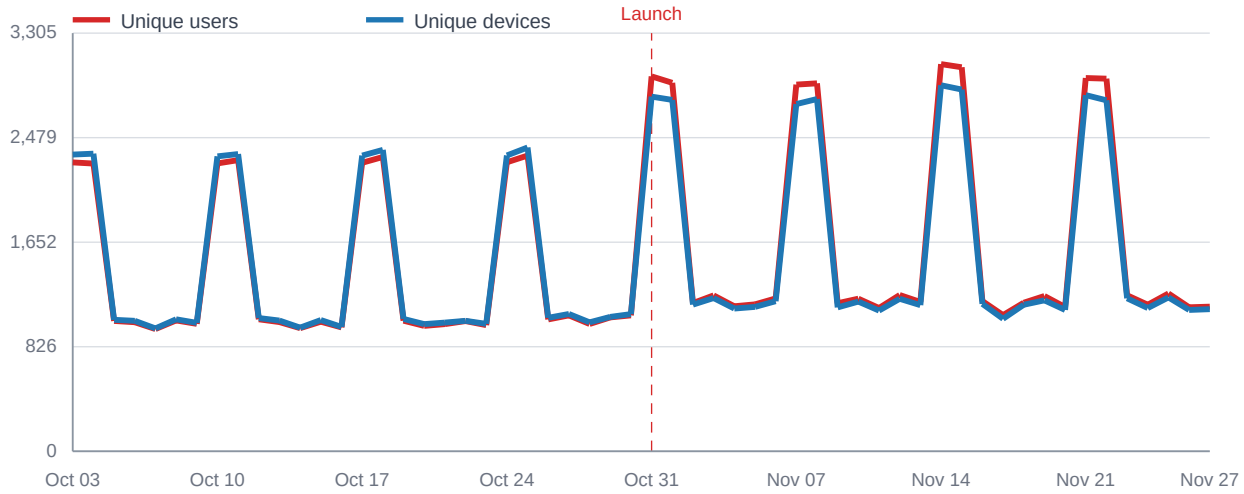
Post-launch segment	Transactions	Users	Revenue	AOV
Non-referral	21,862	8,006	$1,022,599	$46.78
Referral	28,138	10,391	$1,321,404	$46.96

Using the cleaned user-level referral flag, referred users generated $1,321,404 in post-launch revenue. The raw transaction-level flag shows 28,017 referral transactions, which is useful directionally but not clean enough for user-level attribution.

Data Quality and Fraud Signals

The raw data contains 5,857 users who appear as both referred and non-referred. That should not happen if referral status is meant to describe how a user was acquired. For analysis, each user's first observed referral value was assigned to all of that user's transactions.

Unique Devices vs. Unique Users



The device pattern is the biggest warning sign. Before launch, unique devices slightly exceed unique users, which is plausible because some customers use multiple devices. After launch, users exceed devices, and many devices are linked to multiple users.

Metric	Avg daily users	Avg daily devices	Users per device	Multi-user devices	Max users/device
Before launch	1,384	1,408	1.01	0.7%	2
After launch	1,687	1,627	1.37	26.4%	6

This pattern is consistent with users creating additional accounts or sharing devices to capture referral incentives. It does not prove fraud by itself, but it is strong enough to require product and logging investigation before scaling the program.

Country-Level Readout

Performance is uneven across countries. France, the United Kingdom, Italy, Mexico, and Spain show the strongest revenue lift. Germany and Switzerland declined overall despite referral revenue, so they should be diagnosed before additional investment.

Country	Revenue change	Transaction change	Referral revenue
FR	$90,993	+1,724	$209,492
UK	$83,521	+693	$229,122
IT	$71,172	+1,318	$159,965
MX	$57,617	+1,051	$112,240
ES	$56,334	+1,061	$136,397
US	$32,380	-528	$208,676
CA	$11,536	-480	$122,730
CH	$-14,348	-273	$16,340
DE	$-51,598	-1,907	$126,442

Recommended Next Steps

- Keep the referral program running cautiously, but do not scale the incentive until incremental impact is validated.
- Run a controlled test. Ideally, use comparable markets or randomized users with one group eligible for referrals and one holdout group not eligible.
- Instrument the full funnel: invite sent, invite accepted, signup, first purchase, repeat purchase, credit issued, and credit redeemed.
- Add fraud controls: block self-referrals, flag repeated devices, require stronger identity checks for reward eligibility, and delay credit until purchase validity is confirmed.
- Measure margin-adjusted payback and referred-user retention. Revenue lift alone is not enough to decide whether the $10 credit is profitable.
- Investigate countries with weak or negative overall lift, especially Germany and Switzerland, before increasing promotion there.

Conclusion

The referral program appears promising: users and revenue increased after launch, and referred users contributed substantial post-launch revenue. However, the same data also suggests cannibalization and possible abuse, so the program's true causal impact remains uncertain. The best next move is to redesign measurement around a controlled test while improving referral logging and fraud prevention.